Ping is a basic network troubleshooting command used to check whether one device can communicate with another over a network. It also measures how long it takes for data to travel between the devices.

How Ping Works

1. Your computer sends an **ICMP Echo Request** packet to a target device (such as a server or another computer).
2. If the target is reachable, it sends back an **ICMP Echo Reply**.
3. Ping reports:
    - Whether the destination is reachable.
    - The **round-trip time (RTT)** in milliseconds.
    - Whether any packets were lost.

Example Command

Windows, Linux, or macOS:

ping google.com

Example output:

Pinging google.com [142.250.183.14] with 32 bytes of data:

Reply from 142.250.183.14: bytes=32 time=25ms TTL=117

Reply from 142.250.183.14: bytes=32 time=24ms TTL=117

Reply from 142.250.183.14: bytes=32 time=26ms TTL=117

Reply from 142.250.183.14: bytes=32 time=24ms TTL=117

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Important Terms

- **ICMP (Internet Control Message Protocol):** The protocol used by ping.
- **RTT (Round-Trip Time):** Time taken for a packet to go to the destination and back.

- **Packet Loss:** The percentage of packets that never reach the destination or whose replies never return.
- **TTL (Time To Live):** Limits how many routers a packet can pass through before being discarded.

Common Uses

- Check if a device is online.
- Test network connectivity.
- Measure network latency.
- Detect packet loss.
- Help troubleshoot network problems.

Possible Results

- **Successful Ping:** The destination is reachable and replies are received.
- **Request Timed Out:** No reply was received. The device may be offline, the network may have a problem, or a firewall may be blocking ICMP.
- **Destination Host Unreachable:** Your computer or a router cannot find a route to the destination.

Interview Question

Q: What is the purpose of the ping command?

Answer: The ping command is used to verify network connectivity between two devices. It sends ICMP Echo Request packets to a destination and waits for ICMP Echo Reply packets, helping measure response time, detect packet loss, and confirm whether the destination is reachable.